Predicting JEE Chapter Weightage Using Historical Data and Machine Learning

Anirudh, Akshith Anoop, Achyuth Anil Kumar, Kunal K
School of Computer Science and Engineering
RV University
Bangalore, India
1RVU23CSE058, 1RVU23CSE038, 1RVU23CSE019, 1RVU23CSE229

***Abstract*—JEE is one of the toughest engineering entrance exams in India. Students need smart preparation to succeed. We studied JEE papers from 2002 to 2024 to predict which chapters will get more marks in the next exam. We tested Random Forest, SVM, and other ML models to find patterns. Random Forest performed best with a Test RMSE of 1.6508. Our results provide a breakdown of expected chapter marks for JEE 2025 across Physics, Chemistry, and Mathematics. This helps students focus their study time on high-value chapters. The research also shows how JEE patterns have changed over the years.**

***Index Terms*—JEE, Chapter Weightage, Machine Learning, Random Forest, Test Preparation, Study Planning**

I. Introduction

JEE is very difficult. It opens doors to premier institutions like IITs and NITs. Each year, more than a million students compete for limited seats. Smart study plans are as important as understanding the concepts. Past papers show that some chapters consistently get more questions in Physics, Chemistry, and Mathematics. Yet this information is not clearly shared with students. Many end up with ineffective study plans based on uncertain information.

Traditional methods for predicting important chapters rely on teacher opinions or simple counting from past papers. These methods miss the complex patterns of how chapter importance changes over time. With better data tools now available, we can create more accurate prediction models.

Our project tests different ML models to predict chapter weightage for JEE. We used data spanning from 2002 to 2024. Our dataset contains detailed chapter marks for all three subjects over more than twenty years. This gives us a solid foundation to identify patterns and predict future trends. We compared various ML approaches to find the best model based on accuracy, interpretability, and stability. This research could transform JEE preparation by providing data-driven insights instead of guesswork.

II. Related Work

Previous research in exam pattern prediction ranges from basic statistics to advanced ML methods. Here we review key studies that informed our approach.

A. Educational Data Mining for Exam Patterns

Educational Data Mining (EDM) has proven useful for extracting insights from academic data. Algarni [1] applied EDM to predict future exam questions by analyzing student responses and past papers. They used regression and classification models to identify topics likely to reappear. Their results showed that EDM can predict exam content based on historical trends and student performance. This provided a framework for our JEE analysis.

Our project extends this work by using a longer time range (2002-2024) and focusing on JEE-specific characteristics. Unlike Algarni who mainly predicted questions, we focus on chapter weightage across three subjects to enable better study planning.

B. NLP and Automated Question Generation

In question analysis, Nwafor and Onyenwe [2] explored using NLP to automatically create multiple-choice questions from course materials. They identified key concepts from syllabus and past exams to generate questions. Using syntax parsing and other NLP techniques, they demonstrated it's possible to create questions by analyzing curriculum content.

While we aren't generating questions, their work reveals connections between syllabus and question patterns. Their approach to identifying key concepts helped shape our feature engineering, especially in understanding chapter relationships in JEE.

C. ML Model Interpretability in Education

When using ML in education, model clarity matters. Carvalho et al. [3] surveyed methods for explaining ML models, noting that transparent AI models (XAI) are important in education. They found that while "black box" models can be accurate, they are often too complex to understand, causing people to question the results.

This influenced our model selection and evaluation criteria. We wanted both accuracy and clarity in our JEE chapter prediction models. By improving model interpretability, we created predictions that students and teachers can trust and use.

D. Research Gap and Our Contribution

While there is substantial work on educational data mining and exam pattern prediction, a gap exists for competitive exams like JEE. Most research addresses general question prediction without considering the specific challenges of competitive exams and their evolving patterns. Our work addresses this gap by focusing on recent patterns while incorporating historical context.

III. Methodology

Here is our approach to predicting JEE chapter weightage, from data collection through model building and testing.

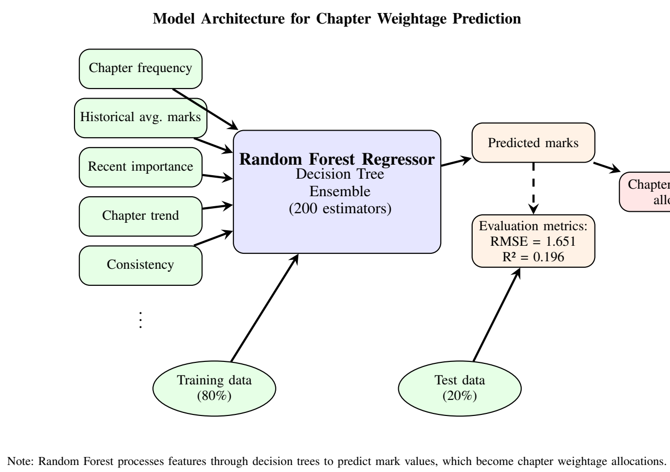

Fig. 1. Model architecture for JEE chapter weightage prediction

A. Data Collection

We compiled JEE exam questions from 2002 to 2024. Our dataset contains 1,632 data points. Each entry shows the year, chapter, subject (Physics, Chemistry, or Mathematics), and marks allocated. This gives us over 20 years of exam patterns to analyze.

We collected data from official JEE papers, academic sources, and public datasets. We started with over 13,000 question records, then filtered them for consistency. The final dataset has a balanced distribution: 584 entries for Mathematics, 546 for Chemistry, and 502 for Physics.

B. Data Preprocessing

We prepared the raw data through several steps:

1) Data Cleaning: We performed thorough cleaning:

- Removed duplicates to prevent bias
- Standardized chapter names across years
- Verified mark allocations against original papers
- Checked for missing values (our final dataset had none)

2) Temporal Segmentation: Recent exams are more relevant for future predictions, so we divided the data:

- Recent years (2015-2024): Given higher weight
- Earlier years (2002-2014): Provided context but with reduced influence

We assigned 75% weight to recent patterns and 25% to historical patterns.

3) Categorical Encoding: For ML model compatibility, we transformed categorical variables:

- Text features (chapter and subject) were one-hot encoded
- We kept original categorical values for easier analysis

C. Feature Engineering

We created informative features to capture different aspects of chapter importance:

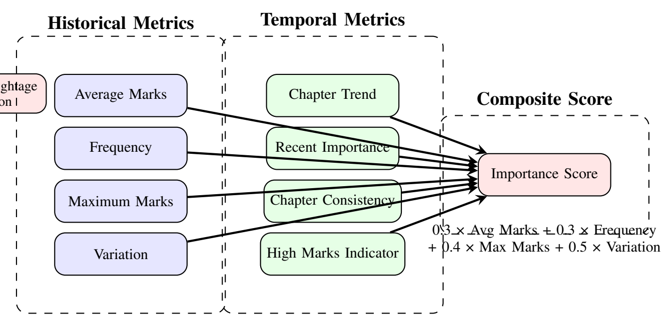

Fig. 2. Feature engineering framework for chapter importance calculation

1) Historical Importance Metrics: For each chapter, we calculated:

- **Average Marks**: Mean marks allocated to the chapter across all years
- **Frequency**: How often the chapter appears in examinations
- **Maximum Marks**: The highest marks ever allocated to the chapter
- **Variation**: Standard deviation of marks, showing volatility

2) Temporal Trend Features: To capture changes in chapter importance over time:

- **Chapter Trend**: Whether marks are increasing or decreasing over years
- **Recent Importance**: Average marks in recent years (2015-2024)
- **Chapter Consistency**: Proportion of years the chapter appeared

3) Derived Indicators: Additional features we created:

- **High Marks Indicator**: Whether the chapter ever received 8+ marks
- **Importance Score**: Combined measure using this formula:

$$\text{Importance Score} = 0.3\times\text{Avg Marks}+0.3\times\text{Frequency}+0.4\times\text{Max M} \tag{1}$$

D. Model Development

We implemented and compared four regression models:

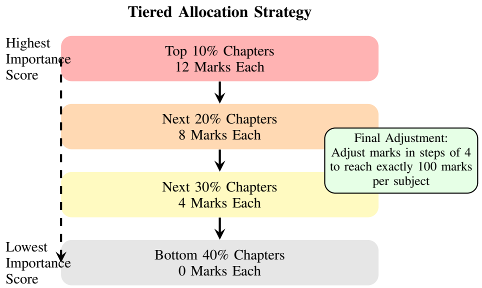

Fig. 3. Tiered chapter weightage allocation strategy

1) Model Selection: We chose diverse models to enable thorough comparison:

- **Random Forest**: An ensemble method using multiple decision trees
- **Support Vector Machine (SVM)**: Effective for complex patterns
- **Gradient Boosting**: Builds models sequentially to correct errors
- **XGBoost**: An optimized implementation of gradient boosting

2) Preprocessing Pipeline: We created a robust data pipeline:

- **Numeric Features**: Applied mean imputation and standardization
- **Categorical Features**: Applied most frequent value imputation and encoding

3) Model Training and Hyperparameter Tuning: Models were trained with these parameters:

- **Random Forest**: 200 estimators, max depth of 15, min samples split of 5
- **SVM**: RBF kernel, C=10.0, epsilon=0.1, gamma='scale'
- **Gradient Boosting**: 200 estimators, learning rate of 0.1, max depth of 5
- **XGBoost**: 200 estimators, learning rate of 0.1, max depth of 5

E. Evaluation Framework

We used comprehensive testing to ensure model reliability:

1) Cross-Validation: We employed 5-fold cross-validation with shuffling to get reliable performance estimates and reduce overfitting.

2) Performance Metrics: Multiple metrics were used to evaluate models:

- **Root Mean Squared Error (RMSE)**: Primary accuracy metric
- **Mean Absolute Error (MAE)**: For average error magnitude
- **R-squared (R^2)**: To measure explained variance
- **Prediction Stability**: Standard deviation of CV scores

3) Classification Evaluation: To assess prediction of importance levels:

- We categorized marks into three levels: low (4), medium (¿4 and 8), and high (¿8)
- Generated confusion matrices to visualize performance
- Calculated precision, recall, and F1-scores

F. Chapter Weightage Allocation Strategy

Based on our analysis and predictions, we created a tiered system to assign marks:

1) Tiered Allocation Approach: Chapters were ranked by importance and allocated marks as follows:

- Top 10% of chapters: 12 marks each
- Next 20% of chapters: 8 marks each
- Next 30% of chapters: 4 marks each
- Bottom 40% of chapters: 0 marks

2) Balancing Mechanism: To ensure each subject totals exactly 100 marks:

- If total exceeds 100: Reduce marks from less important chapters
- If total is below 100: Add marks to more important chapters
- Adjustments made in 4-mark increments to match JEE format

3) Domain-Specific Allocation: We performed allocation separately for each subject:

- Each subject maintains exactly 100 marks
- Subject-specific patterns are preserved
- Allocation reflects both historical and recent patterns

This method combines data analysis with domain knowledge to create a reliable framework for predicting JEE chapter weightage. By integrating historical analysis with ML techniques, we provide insights into chapter importance to guide JEE preparation.

IV. Results and Analysis

Our comparison showed Random Forest performed best. Table I shows performance metrics for all tested models.

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	**Mean RMSE**	**Std RMSE**	**Test RMSE**	**Test R^2**	**Test MAE**
Random Forest	1.669	0.080	1.651	0.196	0.938
SVM	1.766	0.095	1.671	0.176	0.805
Gradient Boosting	1.775	0.048	1.768	0.077	0.977
XGBoost	1.751	0.087	1.708	0.140	0.986

TABLE II
CONFUSION MATRIX FOR RANDOM FOREST MODEL

Actual/Pred.	**Low**	**Med.**	**High**
Low	114	141	3
Medium	0	59	6
High	0	4	0

Random Forest achieved the lowest Test RMSE (1.651) and highest R^2 (0.196) among all models. The confusion matrix shows good performance for medium-importance chapters (91% accuracy) but lower accuracy for low-importance (44%) and high-importance chapters. This likely stems from the imbalanced distribution in the historical data, with fewer high-mark chapters available for training.

Despite these limitations, the model successfully identified key patterns in chapter importance and enabled us to create a practical weightage allocation system. Our final prediction assigns exactly 100 marks per subject, with each chapter receiving 0, 4, 8, or 12 marks based on its predicted importance.

V. Conclusion

This research presents a data-driven approach to predicting chapter weightage for JEE examinations using machine learning. By analyzing historical patterns from 2002 to 2024, we developed a framework that provides insights into chapter importance across Physics, Chemistry, and Mathematics.

Our findings show that ensemble learning methods, particularly Random Forest, effectively capture the complex relationships in chapter importance. The model processes multiple features to identify patterns that might be missed in traditional analysis.

The tiered allocation strategy derived from our predictions offers practical benefits for JEE aspirants:

- Strategic focus on high-weightage chapters for efficient study time use
- Subject-specific prioritization across all three domains
- Balanced preparation considering both historical and recent patterns

While our model shows promising results, we acknowledge limitations in accuracy, especially for high-importance chapters. Future research could address these by adding features like conceptual relationships between chapters, question complexity metrics, and syllabus changes over time.

This research contributes to educational data mining by showing how machine learning can provide valuable insights for exam preparation. By making chapter weightage predictions more objective and data-driven, we help JEE aspirants optimize their study strategies and improve their chances in this competitive examination.

References

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